BIO 005 · WEEK 2 · WORKSHEET

Reading a physiology graph.

Three figures. For each one you work out what it actually says, and just as importantly what it does not say. Print it, do it by hand, then photograph or scan it and attach it to your Week 2 discussion post.

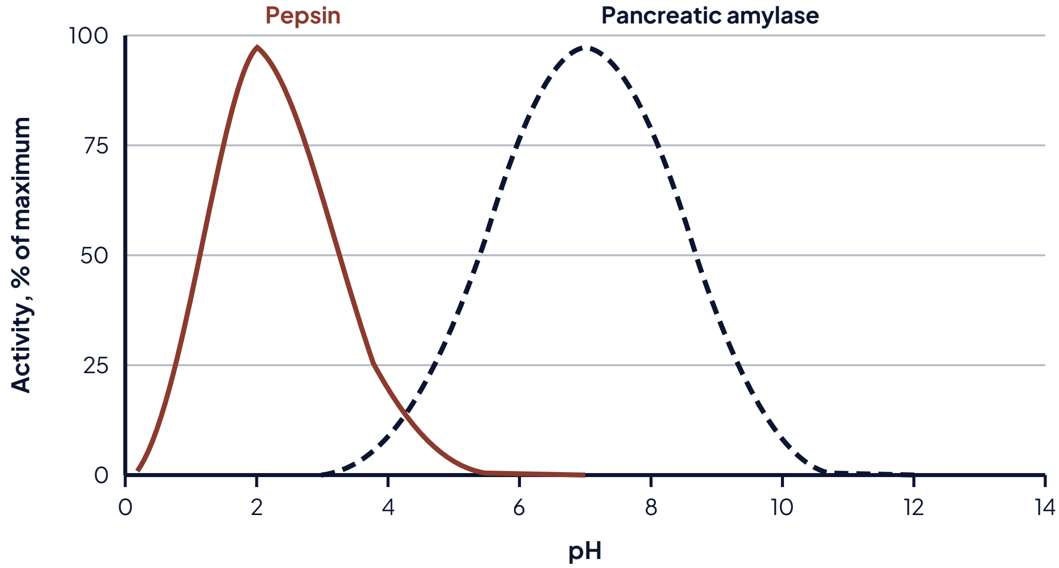
YOUR NAME

DATE

Work it by hand on this printed sheet, in pen or pencil. Do the figures in order and do not look ahead. For each one, glance at it and write what you think it says before you study the axes; your discussion post asks about that first impression, so be honest rather than tidying it up afterwards. Every figure here is real physiology, so these shapes come back later in the term.

FIGURE A

Enzyme activity against pH, for two digestive enzymes



1. Before you study it: in one line, what do you think this figure says?

2. Now the axes. What is plotted on each one, and in what units?

3. Each curve peaks somewhere different. Name the peak pH for each enzyme, and say where in the body each one does its work.

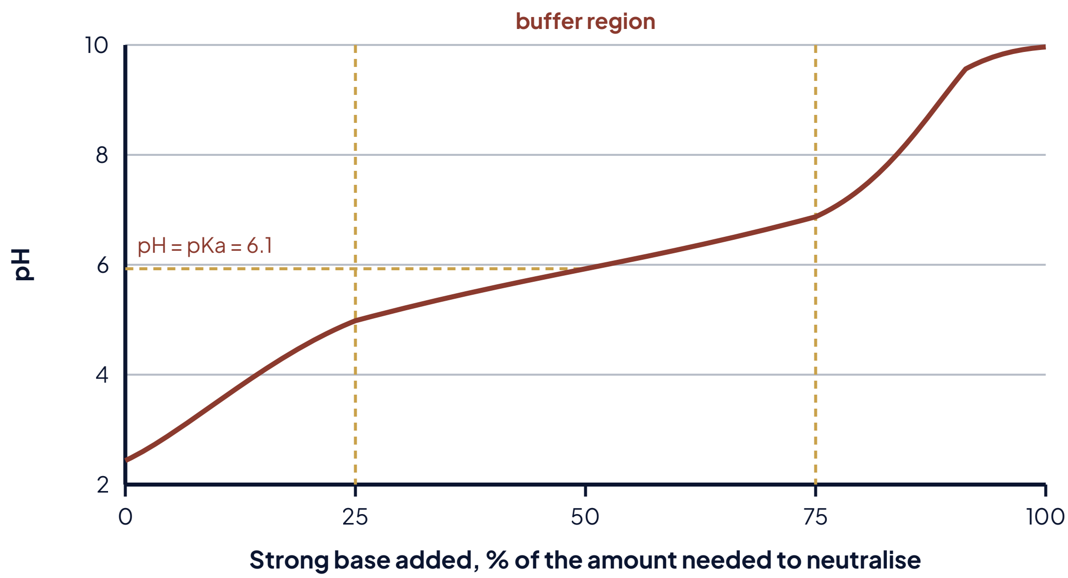
4. Why does activity fall away on BOTH sides of the peak? Answer in terms of the enzyme, not in terms of the graph.

5. What does this figure NOT tell you that a reader might assume it does? Name one thing.

6. A patient takes a drug that raises stomach pH to 5. Using only this figure, what would you predict, and what would you need to know before you trusted that prediction?

FIGURE B

Titration of a weak acid, with the buffer region marked



1. Before you study it: in one line, what do you think this figure says?

2. What is on each axis, and in what units? One axis has no units. Say why not.

3. The middle of the curve is nearly flat. What is physically happening in the solution over that stretch, and why does adding base barely move the pH there?

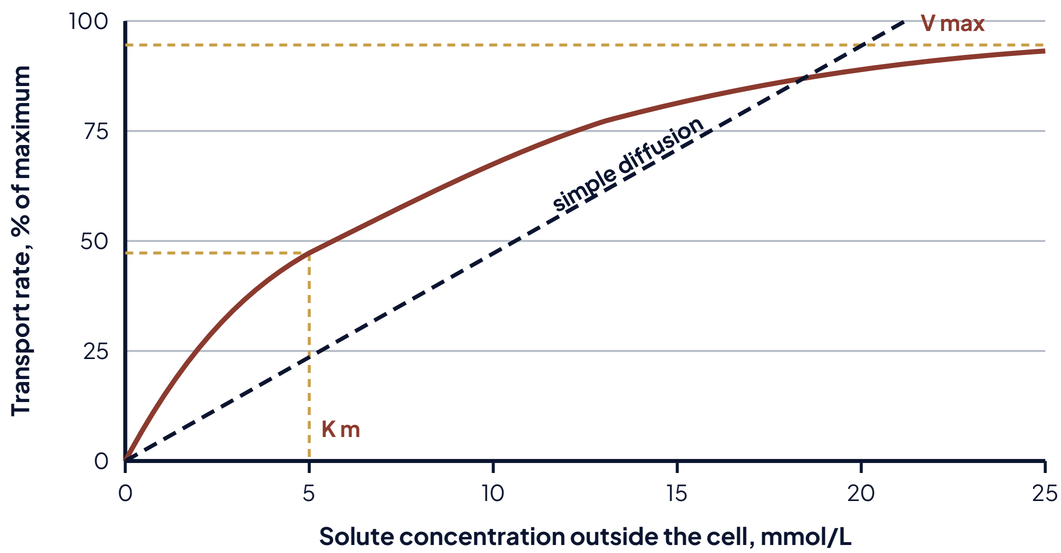
4. The pKa is marked at 6.1. What is true about the ratio of acid to base at exactly that point?

5. 6.1 is the pKa of the carbonic acid and bicarbonate system. Blood is held near pH 7.4, well outside the flat part of this curve. Give one reason that system still works well in the body.

6. What does this figure NOT tell you? Name one thing you would need before using it to reason about a patient.

FIGURE C

Rate of carrier-mediated transport against solute concentration



1. Before you study it: in one line, what do you think this figure says?

2. What is on each axis, and in what units?

3. The solid curve flattens and the dashed line does not. What physical thing about the membrane makes the solid one flatten?

4. Read Km off the graph. What does that value mean, in words a patient could follow?

5. Blood glucose rises above the point where the kidney's carriers saturate. Using the shape of this curve, say what appears in the urine and why.

6. Which of the two lines would still be rising at a concentration off the right edge of this graph, and what does that tell you about that transport route?

LAST ONE, AND IT IS THE ONE THE DISCUSSION ASKS ABOUT

Across all three figures

1. Which figure gave you the most trouble, and what specifically was hard about it? Not "it was confusing." Name the step.

2. Go back to your three first-impression answers. Which one was furthest from what the figure actually says, and what were you assuming that made your first read seem right?

3. One sentence you can reuse: what will you check first, next time you meet a figure you have never seen?

When you are done: photograph or scan every page in order, and attach it to your Week 2 discussion post. The post is due Friday at 10:00 pm; your two replies are due Sunday at 10:00 pm.